

Level: Bachelor
Programme: BE
Course: Object Oriented Software Engineering
Semester: Spring
Year: 2017
Full Marks: 100
Pass Marks: 45
Time: 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

*The figures in the margin indicate full marks.
Attempt all the questions.*

- a) What is the significance of software Engineering? With diagram, explain *scrum* process model for software development.
 - b) What do you mean by software scoping? With example, explain *decision-tree* in order to understand the norms of Make/Buy decision.
 - a) Suppose that Pokhara University is going to develop *Online Learning System* (OLS). You have been asked by the university to mention four (4) major requirements definition for OLS. List the four requirements definition and explain their requirement specifications.
 - b) Draw the Context Level Data Flow Diagram (DFD) and Level 1 DFD for the above OLS.
 - a) What do you understand by software design? Briefly explain different steps that represent a typical task set for component-level design.
 - b) Briefly explain different categories of class. Describe the guidelines for allocating responsibilities to classes.
 - a) Why user interface design is important in software development? Referencing a mobile application for smart agriculture, describe user interface design issues.
 - c) Briefly explain the set of guidelines for formal technical reviews (FTR).
 - a) Considering online shopping software, explain test strategies.
 - b) Define Cyclomatic Complexity (CC). Prove that all methods that calculate CC result the same value for it.
 - a) Define software quality. Explain the activities of software quality assurance (SQA) group for achieving a high-quality end product.
 - b) With diagram, explain software process improvement framework.
- Write short notes on: (Any two)
- a) Types of Software
 - b) Spiral Model
 - c) Software Project Estimation